



Faculty of Nursing

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School Health



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Outlines

- Introduction
- Definitions of school nursing, school health programs, school health team & school health services
- Characteristics of school age period
- Needs of school age group
- Common health problems of school age (young age 6-12 years, adolescent age 13-19 years of age)
- Main cause of morbidity & mortality among school children
- Components of school health programs
 - Health services.
 - Health education.
 - Healthy environment
- Characteristics & importance of healthy school environment
- Classification of Physical school environment:
 - External school environment
 - Internal school environment
- The role of the school nurse

Objectives

At the end of the presentation, the student will be able to:

- Define of school nursing, school health programs, school health team & school health services
- List characteristics of school age period
- Discuss needs of school age group

- Discuss common health problems of school age (young age 6-12 years, adolescent age 13-19 years of age)
- Discuss main cause of morbidity & mortality among school children
- Mention the components of school health programs
- List the characteristics & importance of healthy school environment
- Mention the role of the school nurse

Introduction

Most children are spending more time in school than in any other activity, this makes the school nurse a primary resource in recognizing and managing children's health problems. School health is concerned with health of school children who form a vulnerable of the population who need special school health program because they are constantly growing and developing.

Definitions

- **School nursing** is specialized practices of professional nursing that maintain the well-being, academic success, and life-long achievement of students.
- **School health program** is an important component of community health through the primary responsibility for the health school aged children lives with their parents, their families, and eventually the health of the community.
- **School health team** should include representation from administrator, food service workers, counseling personnel, maintenance workers, medical personnel especially a school nurse and school physician, social workers and teachers
- **School health services** are health services provided by physician, nurses, dentists, health educators, social workers, and teachers to protect and promote health of students and school personnel.

Needs of school age group

1-Nutritional needs:

- Schoolchildren need 2000 calories/ day. 20% protein, 30% fat. 50% CHO, Vitamins, minerals, and water.
- School age children can eat the general family diet with exception of tea and coffee.

- Snacks are most likely to be the primary source of nutrients for the school age children.

2- Protection from infection:

- Booster immunization.
- Healthful school environment.
- Daily observation of all pupils for early detection of any disease or problem.
- Care of contacts.
- Care of absence.
- Screening for infectious diseases as skin disease, scabies.
- Prevention of communicable diseases

School immunization Schedule

Primary level (6-7yrs)

I- BCG check for BCG scar, if scar present not give BCG if no scar give BCG vaccination.

1. OPV Give one booster dose to all at this level.
2. Td Give one dose to all students at this level ' or Give two doses at an interval of 6 to 8 weeks if not vaccinated previously.

Intermediatel level (l2-16 yrs)

- 1- BCG Check for BCG scar if scar present do not give BCG again if no scar give BCG.

- 2- Td Give one booster dose, to all students at this level or if not immunized fully (No record) give two booster doses at an interval of 4 to 6 weeks.

Secondary level Students (17- 18 yrs)

1. OPV Give one booster dose to all at this level.
- TT Give one booster dose if the student is fully immunized in class 6 primary school, or gives two doses at an interval of 4 to 6 weeks if not vaccinated previously

3- Maintenance of health:

- Prevention of accidents.
- A void noise, noisy places around school
- Adequate recreational facilities
- Good ventilation in classrooms.
- Adequate natural and artificial lighting.
- Safe water supply
- Clean toilets and insect control.

4- Exercises

- School must contain a wide play area
- School program should include physical training to improve the circulation of blood and help in muscle flexibility.
- Teach children the ideal body mechanism e.g. proper positioning while sitting, reading, writing etc.

5- Rest and sleep:

- The schoolchild needs adequate periods of rest and sleep. This differs according to the age of the child.

6- Curative services:

- Treatment of any minor ailments
- Referral through health care system.
- Emergency care and first aid.

7- Psycho-social needs:

- School age child needs security, safety, and peer group interaction they need organized games for team play, and learn how to be a leader and follower

8- Health education:

- Good educational, program for prevention of spread of disease. The children must understand the meaning of health and create health. Consciousness, which reflects on their health behavior.

The common health problems of school age (6-12 years) are:

- Accidents
- Respiratory diseases.
- Dental diseases
- Nutritional problems
- School phobia
- Infections

In adolescent age 13 to 19 years are

- Mental and emotional problems
- Menstrual disabilities
- Sexually transmitted diseases
- Other health problems: as sports injuries, dental decay and accidents

Main causes of morbidity and mortality among school children:

1- Morbidity:

- Parasitic diseases.
- Respiratory diseases
- Skin diseases
- Rheumatic fever.
- Infectious diseases
- Dental problems
- Eye diseases
- Accidents & Deficiency disease
- Diarrhea diseases.
- Food poisoning.
- Handicaps
- Emotional problems

2- Mortality:

- **Accidents:** falls, head injury, RTA (Road traffic -An accidents), home accidents... etc.
- **Infectious diseases:** Progressive stages.
- **Rheumatic fever:** Complications

School health program

Objectives of school health program:

- To change beliefs and practices of school children by providing them with adequate knowledge of good healthy habit, which will change their families and community, (child to child, child to family, and child to community)
- To provide a comprehensive school health program that deals with the physical, mental, and social health needs and problems of this population.
- To reduce morbidity and mortality of school age population to improve their quality of life.

Components of school health program:

- Health services.
- Health education.
- Healthy environment



Health services include:

- Screening.
- Case finding.
- Surveillance of immunization status.
- Managing minor complaints.
- Administration of medication.
- Counseling.
- Case management.
- Primary health care.
- Health services for students with special needs.

School Health Education

The children are the parents of tomorrow and if they are educated properly about their health and well - being, then in the future we will have a population that is well informed and have the knowledge to shape

their own lives it is therefore very important that we all work together to inform and help the students to have adequate knowledge for a health Behavior and a healthy attitude.

Objectives:

- To assess and identify the health problems and needs among children requiring health education.
- To educate the students to form a healthy behavior.
- To educate the students of prevention of diseases.
- To educate the student about their well-being.
- To use the student as messengers to convey health back home to their families.

Topics of School Health Education

Personal Health:

- Personal cleanliness.
- Oral health.
- Care of eye, ear, nose, skin, hair and feet.
- Food and nutrition.
- Social and Mental health.
- Safety- accidents- home and traffic (safe bicycle riding).
- Life- style- smoking, alcohol and drugs.

2- Community Health

- Public cleanliness and sanitation.
- Water and health.
- Our living Environment.

3-Health information to students about common health problems among children, first aid, infectious disease, and control of disease.

Healthy environment

School environment: It is the surrounding area of the children at the school and around the school.

Unsafe School environment:

- Considered the main cause of accident as school children are vulnerable group of population and require special healthy environment.

Importance of healthy school environment:

- Promotes optimum psychological & physical growth and developmental among school age children
- Essential to students & staff achievement and develop attitudes & behavior that value healthy life style.

Characteristics of healthy school environment

- Should be safe from physical, chemical, Psychological environmental hazards.
- Comfortable environment has adequate seating, lighting, ventilation, heating, toileting facilities and drinking fountains.
- Organized safety programs.
- Provision of emergency care & adequate transportation
- Safe and sanitary condition of the school.
- School design put into consideration characteristics of the population being served and the needs of disabled students & staff

Classification of healthy school environment

1- Physical environment (Sanitation, health, lighting, safety issue as presence of asbestos or lead in school building)

2- Psychological environment (includes relations between humans inside the school and effects of the community or violence on the safety of children).

Important factors in physical school environment

- quality of air
- Water supply
- Ventilation
- Classroom & its furniture
- School building
- Environmental elements (Tobacco free, drug free & violence free)
- Preparation for first aid and disaster.
- Facilities for hand washing.
- Adherence to standards for heating, cooling, lighting, noise & ventilation

Classification of Physical school environment

1- External school environment

2- Internal school environment

External school environment

Site or school location should be:-

- Should be central to district.
- Easily accessible to the ordinary means of transport.
- Away from noise.
- Take into consideration future expansion of the locality.

- Reasonable land area for yard and playground about 2/3 total area.
- Recommended to have a playing field for out-doors games.
- Street leading to school must be wide enough.
- School should be far from the sources of pollution and should be in good ventilated area.
- School should have more than one enters for the students

Internal school environment

Building

- The intact school building should be designed as the shape (F, U, T, I).
- Certain shape is determined according to the school area.
- School building should have stairs.
- Stairs should be wide enough in relation to the student numbers
- The stair case must be in the level of student's elbow.

Playground

- The important element of internal physical school environment especially for primary school children.
- Each student should have adequate space to play
- The recommended allowance area is about 744 square meters for 200 students
- It means that every student should have at least 3.5- 4 square meters in playground.
- Playground has 1/2 of total school's area.
- Playground contain good part for shelters, with enough seats

Classroom

- Classroom is an important part in the school.
- Classroom must be designed as rectangular shape.
- Dimensions of classroom are 6x8 or 7x5m, height not less than 3.5 m.
- The numbers of the students occupying the class is 24-32 students for the room 8x6 m
- Every student should have 1.5 m in different class shape.
- **Classroom content**
 - Windows**
 - Windows area in classroom must be at least 1/5 or 20% of the floor area and may be more.
 - Windows distributed in one both side walls opposite each other or door, must be kept open all the time.
 - Each window is provided with a hopper on the top.
 - Recommended class room temperature 25-30 c

Lighting

- Lighting sources in the class room are natural and artificial according to time of studying.
- Windows are source of natural lighting so, must go higher up than the-head of the students during sitting.
- Windows glass must be kept always clean and wood casements open.

Noise

- Keeping areas as (shops, playground& music rooms) away from classrooms.
- Keeping the noise at the class room at the level not distributive to others & alerting children to remain quite or speak softly.

Desk & seats

- Designed to keep students back straight .

- Height of the desk depends on age and size of the child
- Desks may be single or doubles
- Good spacing is necessary in the following manner:-
- Distance between front seats and walls should be 2 m at least.
- Aisles between rows of seats are not less than 60 cm.
- Seats should be adequately high.
- Seat is equal $\frac{2}{3}$ the length of the student's thigh.
- It is inclined by a small angle around 15 degree for writing while he reading angle 30.
- The stand for writing should be about 45 cm.

The Board

- It is very essential part in the classroom.
- It should be suitable to size of class for good reading.
- The board paints in classroom must be dark and not glossy to be easily visible when chalk is used.
- The distance between the board and first row should be at least 2 m
- The dimension of the board should be suitable with the class room size at least 1mx1.80 m.
- Small wood stand under the board to protect the students from chalk's dust.

Water supply

- Fountains should be located in accessible position.

Water closets & Urinals

- Self-closing doors
- Ventilation
- General cleanliness

- Insect control.& maintenance
- an slippery floor
- Standard ratio of water closets to female students is 7 water closets per every 100 female students.
- While the ratio to male is 5 water closets and 5 urinals per every 100 male students.
- Should be distributed in whole school building.
- Should be faraway the source of portable water at least 20 m.

Refuse

- Should be collected and disposed every morning.
- Presence of box for wastes in every class, & another big box in playground.
- Garbage should be collected daily from side the school to prevent spread of infection.

Canteen

- Should be always clean, faraway from water closets.
- Clean wall, clean floor.
- Adequate ventilation, good light.
- Source of clean water.
- Fly free.
- Contain Refrigerator, sink,, enough seats & clean cover for food
- Food handlers should be submitted to licensing after routine physical examination.
- Health education includes personnel cleanliness and good health habits.
- The prepared under hygienic condition

Clinic

- School clinic is very important health part in the school environment.
- Clinic should be good ventilated.
- Clinic should be prepared by important equipment.
- Presence of school physical, school nurse.
- Clean water supply.
- Sink to hand washing.
- Adequate light.
- Clinic should be in the first floor.
- Clinic should contain bed with comfortable cover for examination, seats and tables for equipment.
- Blood pressure apparatus, stethoscope.
- Scale for weighting, tape measurement.
- Antiseptic solution, cotton, plaster, bandage, sterile dressing.
- Special equipment for first aid, Para van, wheel chair

Role of the school nurse in school health program:-

a) Healthful school environment

1-Physical environment:-

* The nurse must link with the staff in ensuring that there is a high standard of environmental hygiene (water, sanitation, latrines)

*Daily rounds in the classes to be sure that there is good ventilation, good lighting.

*Daily round in the bathrooms to ensure adequate water supply and proper sewage disposal.

Write a report on the building in a special book and present it to administrator and the physician.

*Be alert to environmental deficiencies and hazards and make suggestions to maintain the department.

2- Non-physical environment:-

*The nurse must involve teachers and parents in children's problems

*She must give support and advice to any child needing it (e.g. listening to adolescents with their problems)

*Promote good teacher pupil relationship.

School health services:-

1-Preventive activities:-

A)-Health appraisal:-

- 1- Taking history (personal, family, medical history)
- 2- Health observation signs of sickness and deviation from normal
- 3- Prepare pupil to screening tests and laboratory investigations
- 4-Screen vision, hearing and measure weight and height
- 5- Help in laboratory investigations
- 6- Assist in comprehensive medical examination for schoolchildren
- 7-Interpret the findings of screening tests and laboratory investigation to the teacher and parent
- 8-Conduct special surveys to detect any health problem

b) Follow up and counseling for the children deviated from normal.

c) Prevention and control of communicable disease:-

- 1- Provide immunization
- 2- Daily observation of pupils for early case findings, referral and isolation.
- 3- Prevention and control of outbreaks of infections.
- 4- Period of isolation of the child at home for each of the following diseases:-

Hepatitis21days

*Whooping cough18 days

*Measles.....14 days

*Chicken pox.....14days

*Mumps.....14 days

*Scarlet fever.....10 days

*G. Measles7 days

4-Provide health teaching about period of isolation.

5- Care of contacts.

6- care of absence.

7- Be sure that food handlers in the school pass through physical examination and have health certificate.

8- Insure healthful school environment.

D) Early case finding and referral of non-communicable diseases and parasitic disease.

E) Emergency care and first aid:-

The nurse is responsible to carry out first aid treatment for the injured due to accident, and if necessary transfer them to hospital or clinics.

- Be sure that there is enough first aid equipment and supplies in the school to carry out first aid.

*The nurse should arrange for in service training program for teacher and interested in first aid.

F- care of handicapped children:-* Identify child with handicapped problem (e.g., hearing problem, or vision, epilepsy, mobility difficulties) and arrange referral, treatment and support.

*Help handicapped students to accommodate with their defects .

H- Health serviced to the school personnel according to their needs

2- Curative activities:

- 1- Referral for treatment of any discovered diseases.
- 2-Children in need of medical treatment or special care should be identified (as child with learning problems)
- 3- Provide medication for chronically ill children.
- 4-Felling health records and keeping it.
- 5-Interpret the finding and recording the progress in special record.
- 6-Keep the health records available to teachers and physicians.

c- School health education:

- 1- The nurse can participate in the parent council and stimulate the interest and cooperation of parent in the school health program.
- 2- Formulate health association.
- 3- Conduct health conferences with teachers about the student's health problem.
- 4- Assist in planning curriculum for health instruction
- 5- Emphasize utilization of community recourses and guide parents in selecting suitable agencies for assistance of their children.
- 6- Give health education to sick children about how to prevent the re infection.
- 7-The nurse can participate in class discussion in occasions such as immunization, medical examination and accidents.
- 8- The nurse should help in planning and conducting school health projects.